Performance and Scaling

General scaling properties of uG4

Our base code uG4 is parallelized using MPI and written in C++ and has been extensively benchmarked (Vogel et al., 2013; Heppner et al., 2013) and shown to scale beyond 200,000 processes. In order to test the code on the SDSC Comet supercomputer, we ran a test problem using the Laplace equation on a three-dimensional unit cube. The second order elliptic partial differential equation reads

$$\Delta\Phi = 0, \tag{1}$$

where Φ is a scalar function in the domain $\Omega \subset \mathbf{R}^n$, and Δ is the Laplace operator. Strong scaling is demonstrated for a domain consisting of 16,974,593 degrees of freedom, see Tab. **??** and Fig. **??**. The execution time roughly halves when doubling the number of processes. A weak scaling study, see Tab. **??** and Fig. **??** on SDSC Comet. With each refinement of the grid the degrees of freedom (DoFs) increase by a factor of 8 and thus we increased the number of processes eightfold in each refinement step.

Strong scaling in uG4			
DoFs	*Runtime [s]*	*# Processes*	*# Nodes*
16,974,593	41	24	1
16,974,593	20	48	2
16,974,593	14	72	3
16,974,593	10	96	4
16,974,593	8	120	5
16,974,593	7	144	6
16,974,593	5	168	7
16,974,593	2	192	8

Table 1: Runtimes of simulations of a *3d Laplace problem* on a fixed unit cube domain (seven regular refinements) with uG4 on the SDSC Comet HPC cluster. Note that each node can allocate at maximum 24 processes and the maximum number of processes or cores per job is limited to 1,728 processes in total.

Weak scaling in uG4			
DoFs	*Runtime [s]*	*# Processes*	*# Refinements*
27	5.06	1	0
216	5.06	8	1
1,726	6.92	64	2
13,824	6.92	512	3
110,592	5.50	4096	4

Table 2: Runtimes of simulations of a *3d Laplace problem* on a unit cube domain (different number of regular refinements) with uG4 on the SDSC Comet HPC cluster. Note that each grid refinement increases the DoFs by a factor 8.

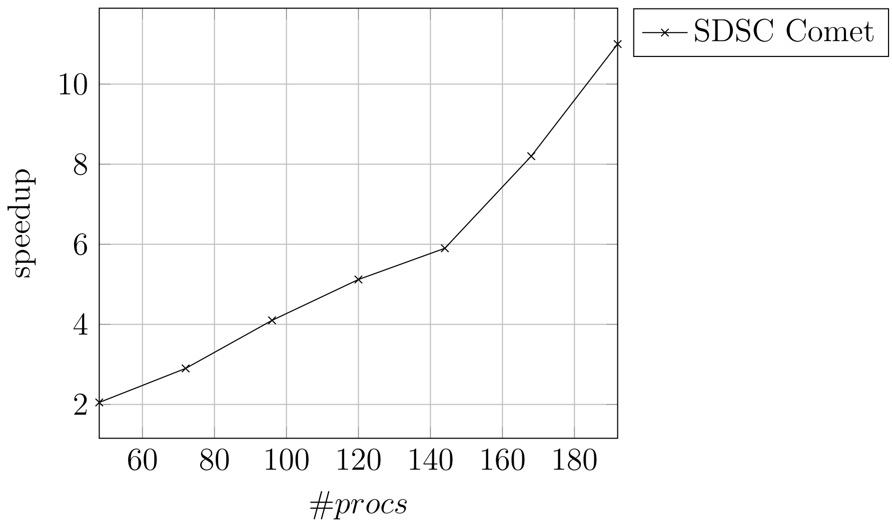


Figure 1: Speedup of distributed execution. #procs denotes the involved processes on the SDSC system Comet.

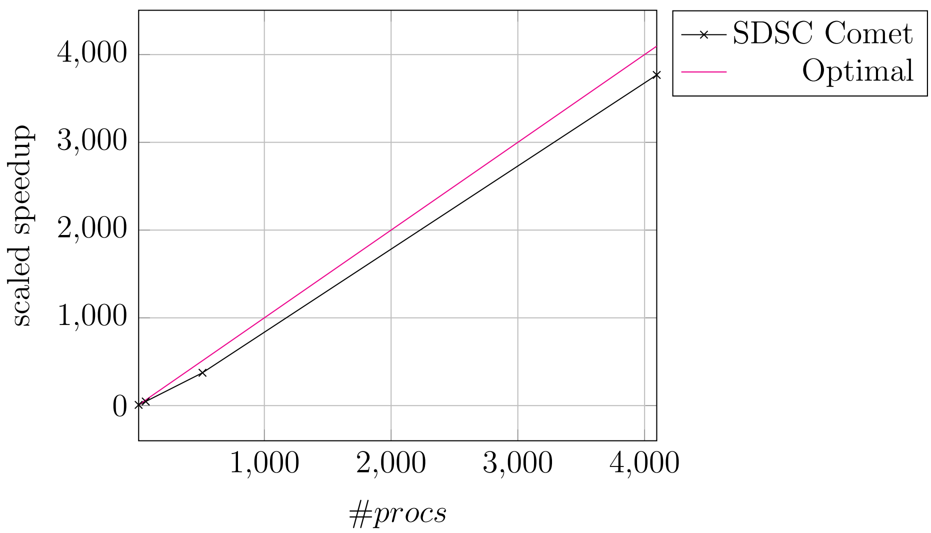
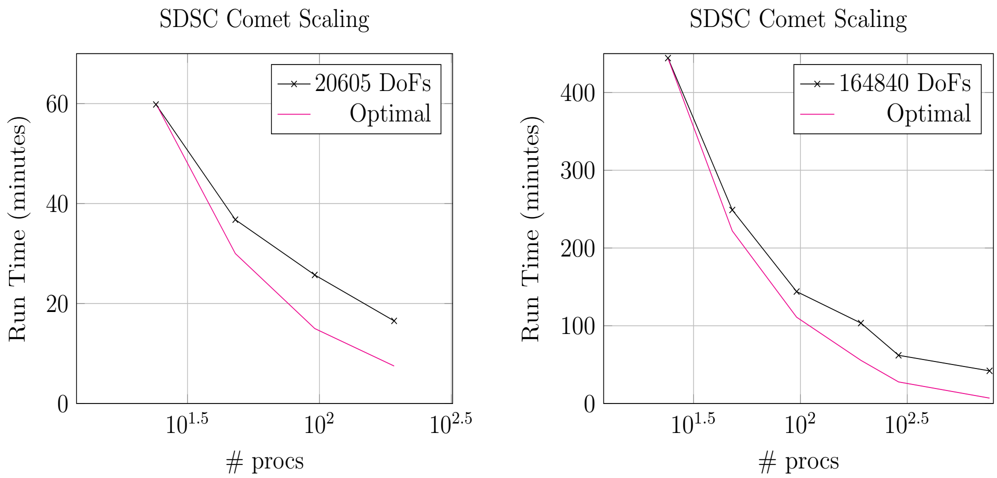


Figure 2: Scaled speedup of distributed execution. #procs denotes the involved processes on the SDSC system Comet. Note the optimal scaling and achieved scaling behaviour of the test problem.

Subproject 1: Dendritic spine calcium simulations

For this subproject we ran simulations with 0 refinements (20,605 DoFs) and 1 refinement (164,840 DoFs) on Spine-Dendrite geometries that contain no ER, with simulation end time 0.02 seconds. Figure ?? demonstrates the strong scaling behavior of the simulations. In particular, the geometry with 164,840 DoFs was used in running a `Baseline run` with Wall time 60 minutes, and simulation consumed 30 SU's when parallelized across 384 processors. Therefore, we anticipate that each `Baseline run` and `RyR run` will use 30 core-hours for a simulation end time of 0.02 seconds. Our resource Table 1 is based on this preliminary run, we scaled accordingly the number of service units for the `Calcium Pulse runs` which require a minimum of 3 seconds of simulation time.



(a) (b)

Figure 3: *Strong Scaling* Here we demonstrate the strong scaling nature of a Spine-Dendrite Calcium simulation using a spine dendrite geometry *without ER*: Figure (a) 20,650 DoFs (0 refinement) and Figure (b) 164,840 DoFs (1 refinement) using the trial XSEDE (Comet) account resources with 0.02 ms end time.

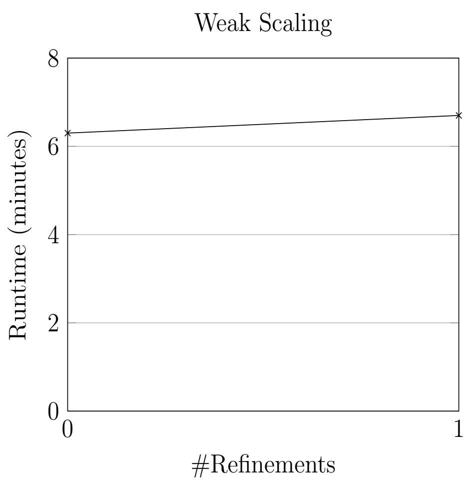


Figure 4: Weak Scaling for Spine-Dendrite simulation runs *without ER* present with respect to Refinement level and runtime. For 0 refinement, 24 procs. were used and for 1 refinement 192 procs. were used.

Subproject 2: Whole-cell calcium simulations

For the whole-cell calcium model a strong and weak scaling study was conducted on SDSC Comet, see Fig. **??** and **??** for runtimes and Table **??** and **??** for the associated problem sizes.

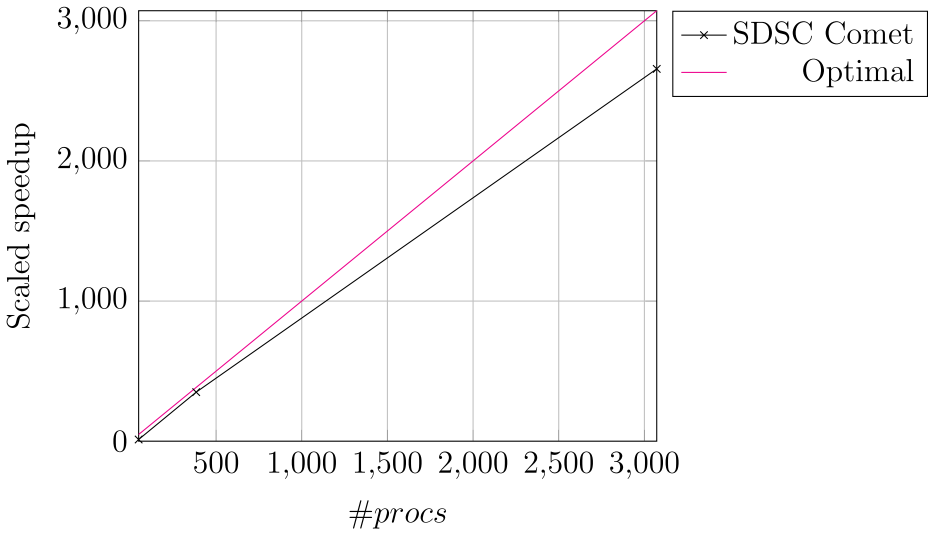


Figure 5: Weak scaling result for the subproject *whole-cell calcium dynamics*. Scaled speedup of distributed execution, where #procs denotes the involved processes on SDSC Comet, shows scaling behavior for whole-cell dynamics. Problem sizes are illustrated in Table **??**.

DoFs	*Runtime [s]*	*# Processes*	*# Refinements*
7,056	278	48	0
56,448	384	346	1
451,485	370	3072	2

Table 3: Weak scaling result for whole cell dynamics, geometry sizes and number of involved processes on SDSC Comet.

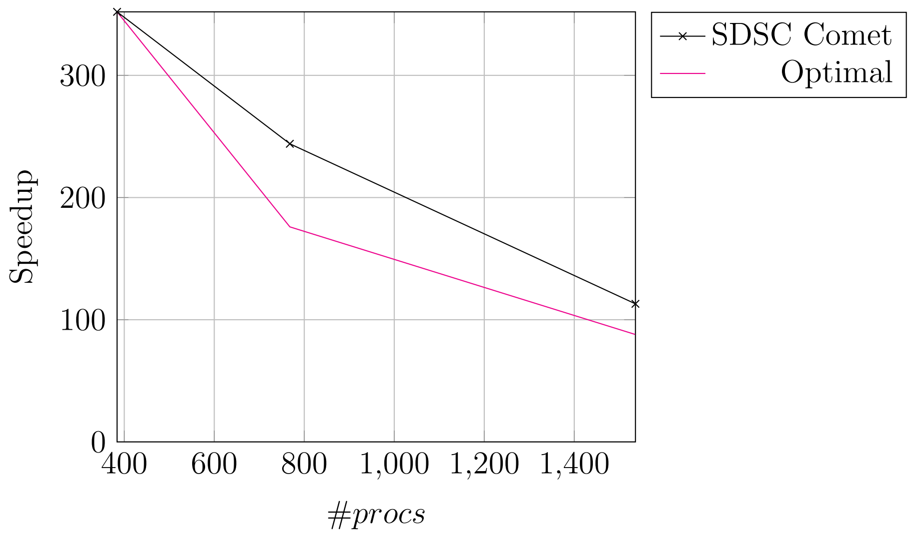


Figure 6: Strong scaling result for the subproject whole-cell calcium dynamics. Scaled speedup of distributed execution, where #procs denotes the involved processes on SDSC Comet, demonstrate scaling behavior. Problem sizes are illustrated in Table **??**.

DoFs	*Runtime [s]*	*# Processes*	*# Nodes*
451,485	352	384	16
451,485	244	768	32
451,485	123	1536	64

Table 4: Strong scaling results fo whole-cell dynamics, geometry sizes and number of involved processes and nodes on SDSC Comet.